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Introduction to DL

SOTA in traditional ML

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Agenda

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**Introduction to
Deep Learning**



02

Data quality



03

Model fairness



04

**Model
Explainability**

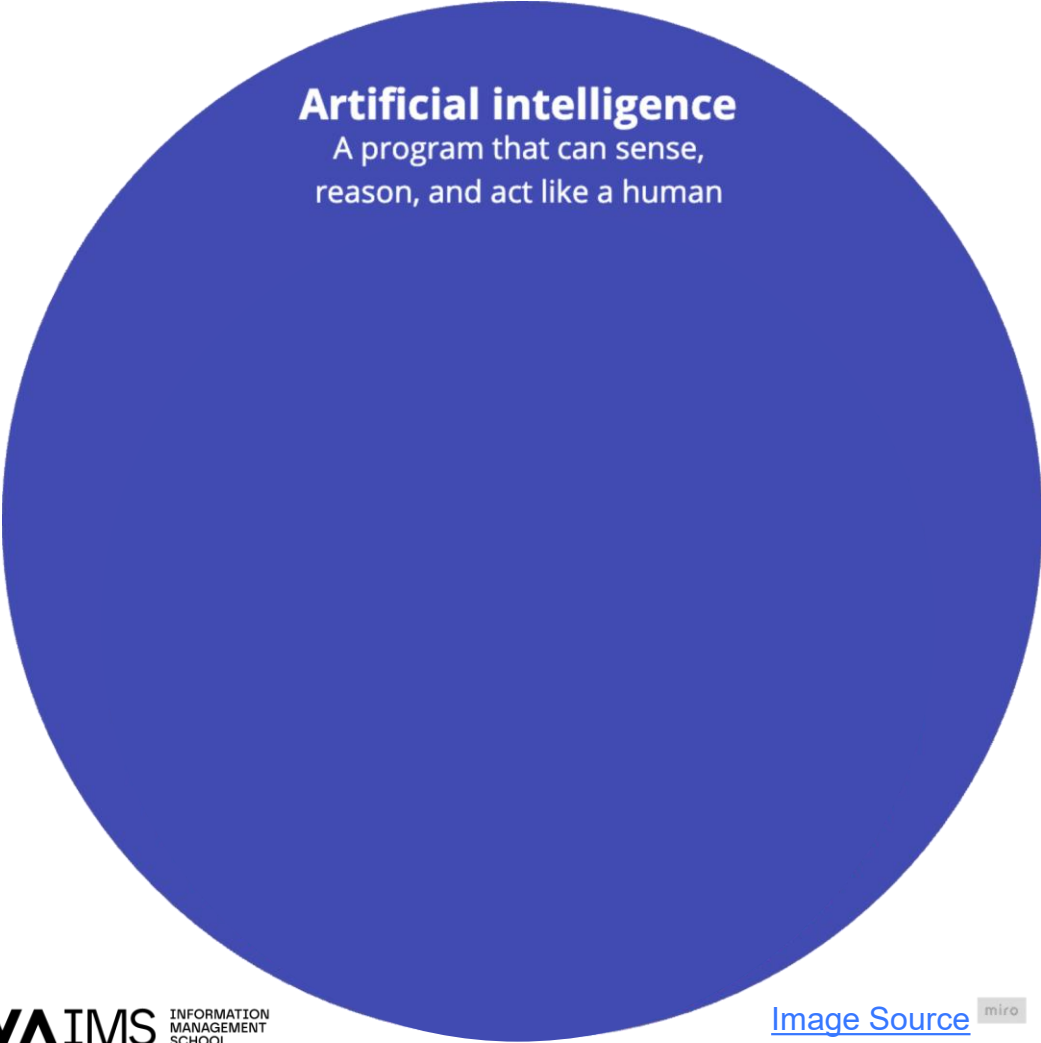


01.

Introduction to Deep Learning

1.1. From traditional ML to Deep learning

Introduction



Artificial intelligence

A program that can sense, reason, and act like a human

Artificial Intelligence (AI)

The broad field of creating systems capable of performing tasks that typically require human intelligence.

Machine Learning (ML)

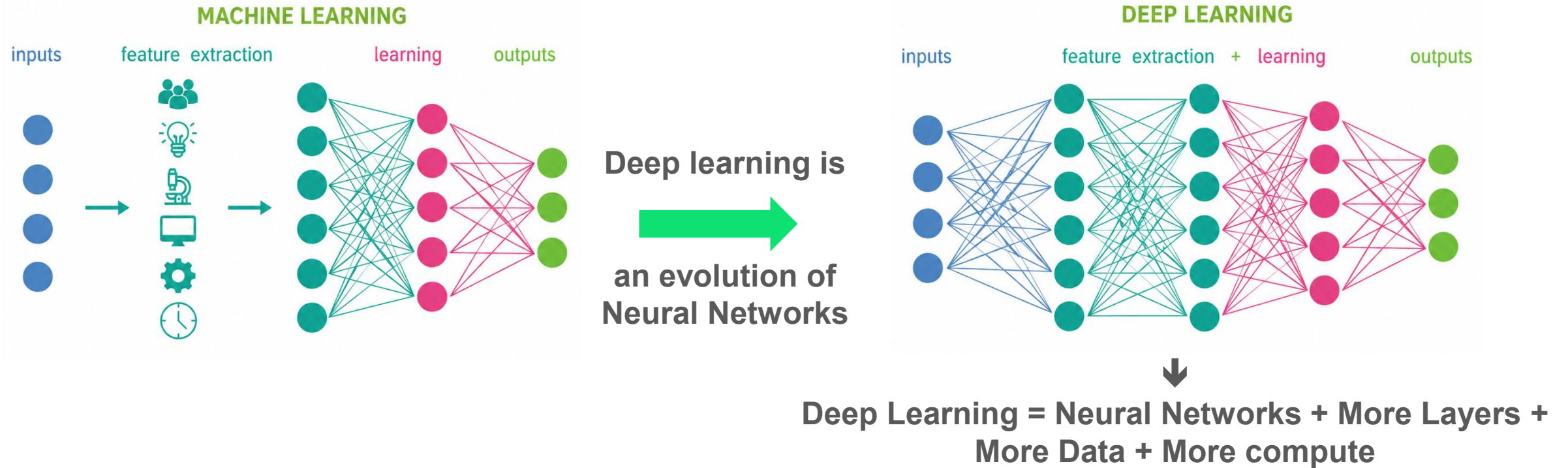
A subset of AI where systems learn patterns from data to make predictions or decisions.

Deep Learning (DL)

A subset of ML that uses multi-layered neural networks to learn complex representations automatically.

1.1. From traditional ML to Deep learning

Introduction



But what are the main differences?
And when should I apply each?

1.2. From traditional ML to Deep learning

What is Traditional Machine Learning?

What is traditional Machine Learning (ML)?

Traditional Machine Learning uses statistical algorithms to learn relationships from structured data.

Characteristics

- Relies heavily on **feature engineering**
- Performs well with **smaller datasets**
- **Easier to interpret and explain**
- **Faster training times**

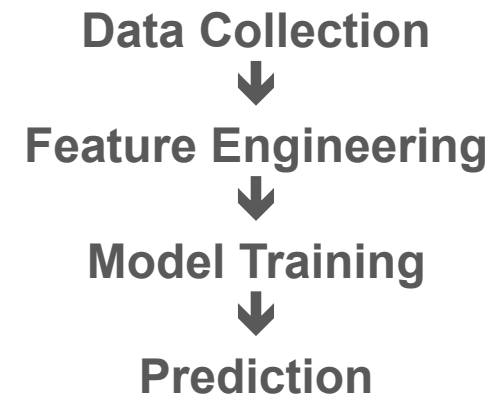
1.2. From traditional ML to Deep learning

What is Traditional Machine Learning?

What is traditional Machine Learning (ML)?

Common Algorithms

- Linear Regression
- Logistic Regression
- Decision Trees
- Random Forests
- Support Vector Machines (SVM)
- ...



Traditional ML depends on human expertise to identify relevant features.

1.3. From traditional ML to Deep learning

What is Deep Learning?

What is Deep Learning (DL)?

Deep Learning uses artificial neural networks with multiple hidden layers to automatically learn features and patterns from data.

Characteristics

- **Learns features automatically**
- **Excels with large datasets**
- **Handles unstructured data effectively**
- **Achieves state-of-the-art performance in many domains**

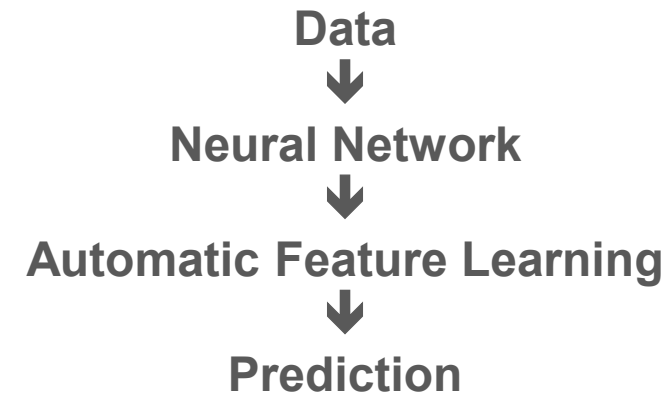
1.3. From traditional ML to Deep learning

What is Deep Learning?

What is Deep Learning (DL)?

Common Algorithms

- CNNs (Convolutional Neural Networks)
- RNNs (Recurrent Neural Networks)
- LSTMs (Long Short Term Memory)
- Transformers
- LLMs (Large Language Models)
- ...



Deep Learning reduces the need for manual feature engineering by learning representations directly from data.

1.4. From traditional ML to Deep learning

Traditional Machine Learning vs Deep Learning

Isn't the traditional Machine learning enough?

Traditional ML is extremely effective for ...

- Tabular data
- Structured business datasets
- Interpretable prediction
- Relatively small datasets

1.4. From traditional ML to Deep learning

Traditional Machine Learning vs Deep Learning

However...

The limitations in ML

Traditional ML still struggles when...

- Features are difficult to engineer
- Data is unstructured
- Context matters strongly

Examples:

- Image recognition, Language understanding, Speech, Video, Multimodal reasoning...

Traditional ML is still dominant in many enterprise scenarios...

Deep learning did not replace everything!

1.4. From traditional ML to Deep learning

Traditional Machine Learning vs Deep Learning

Computational Requirements

Factor	Traditional ML	Deep Learning
Hardware	CPU	GPU/TPU
Training Time	Minutes to hours	Hours to weeks
Memory Usage	Low	High
Energy Consumption	Low	High
Cost	Lower	Higher

1.4. From traditional ML to Deep learning

Traditional Machine Learning vs Deep Learning

Aspect	Traditional ML	Deep Learning
Data Requirement	Small–Medium	Large
Feature Engineering	Manual	Automatic
Interpretability	High	Lower
Computational Cost	Low	High
Training Time	Faster	Slower
Structured Data	Excellent	Good
Unstructured Data	Limited	Excellent
Scalability	Moderate	High

1. From traditional ML to Deep learning

RECAP

- Deep Learning is a subset of Machine Learning
- Traditional ML relies on feature engineering while DL learns features automatically
- ML excels with structured/tabular data while DL excels with images, text, audio, and video
- DL requires more data and computational resources
- There is no universally best model

Choose always the simplest model that solves the problem effectively!



02.

Data quality

2.1. Data quality

Introduction

- Data quality is often more important than model complexity.
 - Poor-quality data creates **biased**, **unstable**, and **unreliable** models.
- 60% to 80% of ML project effort is spent on data preparation.
 - "Garbage In, Garbage Out" remains a fundamental ML principle.
- Data quality directly impacts model generalization.

Most ML failures are data failures, not model failures.

2.1. Data Quality challenges

Introduction

Challenge	Example
Missing Data	Empty customer income
Noisy Data	Sensor errors
Inconsistent Data	Different date formats
Duplicates	Same customer twice
Label Errors	Wrong class assignment
Data Drift	Customer behavior changes
...	...

2.2. Data Quality

Data Centric AI

“Instead of focusing on the code, companies should focus on developing systematic engineering practices for improving data in ways that are reliable, efficient, and systematic.

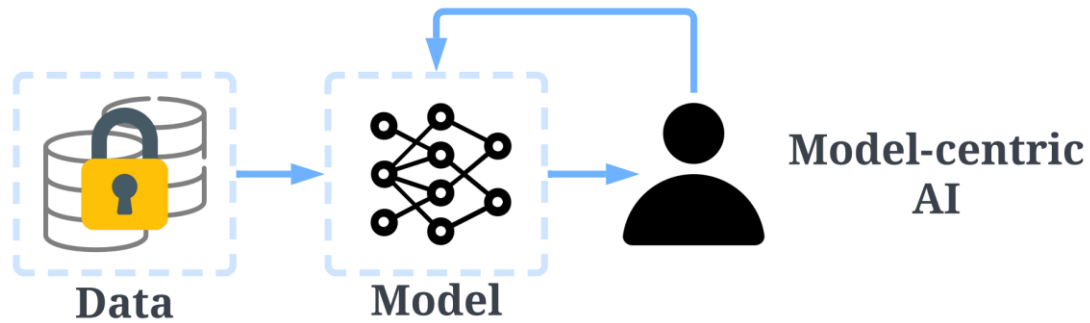
In other words, **companies need to move from a model-centric approach to a data-centric approach.**”

Andrew Ng, AI pioneer, and entrepreneur, known for co-founding Google Brain, Coursera, and DeepLearning.AI



2.2. Data Quality

Data Centric AI



How can we build a better model?

- Try XGBoost instead of Random Forest
- Tune hyperparameters
- Use a larger neural network
- Add more trees to our Random Forest
- ...

2.3. Data Quality challenges

SOTA

How we deal with data quality issues **nowadays?**

PREVENTION | VALIDATION | MONITORING | REMEDIATION

2.3. Data Quality challenges

SOTA

PREVENTION | VALIDATION | MONITORING | REMEDIATION

Stop "bad" data from entering the system

Data Validation at Ingestion

Validate schemas, data types, formats, ranges, and business rules before data enters the system.

- Tools: [Pandera](#), [Great Expectations](#).

Data Testing in Pipelines

Treat data like software by adding automated tests during development and deployment.

- Prevents bad data from propagating downstream.

Data Governance and Standards

Define data ownership, quality metrics, naming conventions, and validation policies.

- Ensures consistent data management across teams.

Prevent bad data from propagating downstream

2.3. Data Quality challenges

SOTA

PREVENTION | **VALIDATION** | MONITORING | REMEDIATION

Check whether data meets defined quality rules!

Data Validation at Ingestion

Validate schemas, data types, formats, ranges, and business rules before data enters the system.

- Tools: [Pandera](#), [Great Expectations](#).

Data Testing in Pipelines

Treat data like software by adding automated tests during development and deployment.

- Prevents bad data from propagating downstream.

Check data against schemas, rules, and expectations

2.3. Data Quality challenges

SOTA

PREVENTION | VALIDATION | **MONITORING** | REMEDIATION

Continuously observe data quality after data is flowing through the system!

Continuous Data Quality Monitoring

Monitor completeness, accuracy, consistency, freshness, and anomalies over time.

- Tools: [Evidently AI](#)

Alerting and Incident Management

Trigger alerts when data quality thresholds are violated.

- Enables rapid investigation and resolution.

Detect issues in production such as drift, missing data, anomalies, and freshness problems.

2.3. Data Quality challenges

SOTA

PREVENTION | VALIDATION | MONITORING | **REMEDIATION**

Fix or respond to identified data quality issues!

Automated Data Cleaning

Handle missing values, duplicates, inconsistent formats, and invalid records automatically.

- Commonly implemented in ETL (Extract, Transform, Load) pipelines.

Alerting and Incident Management (investigation and resolution)

Trigger alerts when data quality thresholds are violated.

- Enables rapid investigation and resolution.

Correct or resolve data quality issues once detected

2.3.1. Data Quality challenges

Automated Data Cleaning

Traditional approach:

- Manual inspection
- Rule-based validation

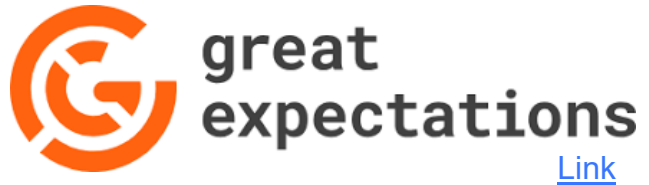
Modern approach:

- Automated anomaly detection
- Schema validation
- Data quality pipelines

2.3.1. Data Quality challenges

Automated Data Cleaning

Examples:



Examples:

- Detect impossible ages
- Invalid categories
- Duplicates
- Outliers
- ...

2.3.2. Data Quality challenges

Missing data

Missing values reduce statistical power!

- Common mechanisms:
 - MCAR (Missing Completely at Random)
 - MAR (Missing at Random)
 - MNAR (Missing Not at Random)
- Can introduce severe bias.
- Particularly problematic in healthcare and finance.

2.3.2. Data Quality challenges

Missing data

Method	Status	Notes
Mean/Median Imputation	Baseline	Fast but often loses information
KNN Imputation	Traditional	Uses similar observations
MICE (Multiple Imputation by Chained Equations)	Strong classical method	Widely used in statistics and healthcare
MissForest	Strong classical method	Random Forest-based, excellent for tabular data
XGBoost / LightGBM Native Handling	Industry standard	Often better than explicit imputation
...		

2. Data Quality challenges

RECAP

- **Data quality is a critical factor** in machine learning success.
 - Poor-quality data leads to **biased, unstable, and unreliable models**.
- Most ML effort is spent on **data preparation and quality management**.
- The principle "**Garbage In, Garbage Out**" highlights the impact of poor data on model performance.
 - High-quality data improves **generalization, reliability, and trustworthiness**.

Most ML failures are caused by data quality issues rather than model limitations.



03.

Model fairness

3.1. Model fairness

Introduction to bias

What is Algorithmic Bias?

Algorithmic bias occurs when a model systematically produces unfair outcomes for certain individuals or groups.

Examples:

- Gender bias in hiring
- Racial bias in lending
- Age bias in insurance

3.1. Model fairness

Introduction to bias

Prior offense

- 1 attempted burglary

Subsequent offenses

- 3 drug possessions

Two Drug Possession Arrests



DYLAN FUGETT

LOW RISK

3



BERNARD PARKER

HIGH RISK

10

Fugett was rated low risk after being arrested with cocaine and marijuana. He was arrested three times on drug charges after that.

Prior offense

- 1 resisting arrest without violence

Subsequent offenses

- None

[Source](#)

3.1. Model fairness

Introduction to bias

It is not necessarily malicious.

- Bias can occur even when everyone, from the data collectors to the engineers to the medical professionals, have the best intentions.

It is not one and done.

- Just because an algorithm has no bias now does not mean it has no potential later.

It is not new.

- Researchers have raised concerns over the last 50 years.

3.2. Model fairness

Source of Bias

Examples

Source	Example
Historical Bias (Past discrimination)	A hiring model trained on past company hires learns to prefer men because historically most managers were male.
Sampling Bias (Underrepresented groups)	A health prediction model trained mostly on data from young adults performs poorly on elderly patients.
Representation Bias (Missing populations)	A facial recognition dataset contains very few images of darker-skinned individuals, leading to lower accuracy for those groups.
...	...

3.3. Model fairness

Why fairness matters?

Avoids Discrimination

- Prevents models from unfairly disadvantaging individuals or groups based on attributes such as gender, race, age, or disability

Builds Trust and Acceptance

- Users are more likely to trust and adopt AI systems that treat people fairly.

Ensures Ethical Decision-Making

- AI should support equitable outcomes, especially in high-impact domains like healthcare, hiring, lending, and criminal justice.

Supports Regulatory Compliance

- Many regulations require organizations to demonstrate that automated decisions are fair and non-discriminatory.

3.4. Model fairness

Metrics on fairness

METRICS

Demographic Parity

Equal Opportunity

Equalized Odds

...

Does the model approve
loans equally across groups?

3.5. Model fairness

Bias mitigation

In pre-processing

Rebalancing

Adjust the dataset so underrepresented groups have more influence.

Dataset: 90% men, 10% women.



Increase the weight of women's observations during training.

Resampling

Change the composition of the training data.

Dataset: 90% men, 10% women.



Oversample minority group examples.
Undersample majority group examples.

The goal: Give all groups similar representation in the training data.

3.5. Model fairness

Bias mitigation

In processing

Fairness constraints

Add fairness requirements directly into model training.

Instead of optimizing only:

$$\text{Loss} = \text{Prediction Error}$$

optimize:

$$\text{Loss} = \text{Prediction Error} + \lambda(\text{Unfairness})$$

The goal: Improve fairness while maintaining accuracy.

3.5. Model fairness

Bias mitigation

In post- processing

Threshold Adjustments

Keep the trained model unchanged but use different decision thresholds.

Loan approval model:

- Group A: approve if score > 0.60
- Group B: approve if score > 0.55

This can help equalize:

- True Positive Rates
- False Positive Rates

The goal: Improve fairness after the model has already been trained.

3. Model fairness

RECAP

- **Algorithmic bias** occurs when a model produces systematically unfair outcomes for certain groups.
- Bias can originate from **data**, **features**, or **model design**.
- Fairness is essential to ensure **equitable, trustworthy, and responsible AI systems**.
 - Unfair models can lead to **discrimination**, loss of trust, and regulatory issues.
- Fairness should be evaluated throughout the ML lifecycle using appropriate metrics and monitoring.

Fairness helps build AI systems that are transparent, ethical, and beneficial for all users.



04. **Model** **interpretability and** **explainability**

4.1. Model Interpretability and Explainability

Introduction

Why model interpretability matters?

Interpretability refers to understanding **how and why** a model makes predictions.

- Essential for **trust**, **transparency**, and **accountability**.

Critical in high-risk domains:

- Healthcare, Finance, Insurance, Government...

Additionally...

- Supports debugging and model improvement.
- Increasingly required by regulations and AI governance frameworks

4.1. Model Interpretability and Explainability

Introduction

Accuracy vs Interpretability trade-off

Interpretable Models (usually less accurate)

Linear regression
Logistic Regression
Decision Trees
Rule-Based Models

Black Boxes Models (usually more accurate)

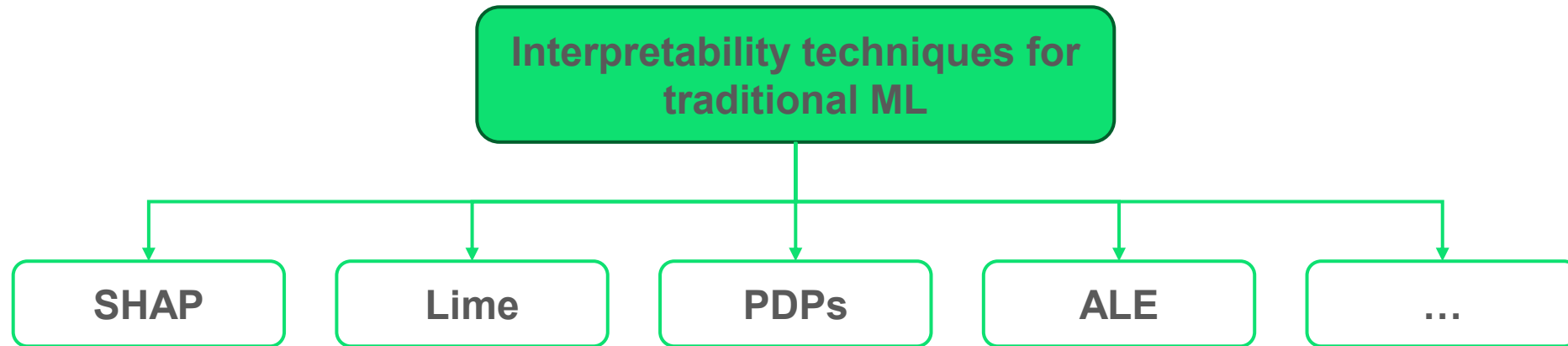
Random Forests
Gradient Boosting
XGBoost
SVM (non-linear kernels)

Simpler models are easier to explain while complex models often achieve higher predictive accuracy.

Balancing transparency and performance is a major challenge.

4.2. Model Interpretability and Explainability

XAI Techniques



Explainable AI (XAI) techniques

Techniques used to understand why a machine learning model makes certain predictions

4.2. Model Interpretability and Explainability

XAI Techniques

SHAP

Assigns each feature a contribution value explaining a specific prediction using Shapley values from game theory. ([Link](#))

LIME

Explains a prediction by fitting a simple interpretable model around that specific data point. ([Link](#))

PDP (Partial Dependence Plot)

Shows the average effect of a feature on the model's predictions. ([Link](#))

ALE (Accumulated Local Effects)

Shows how a feature influences predictions while accounting for correlated features. ([Link](#))

...

4.2. Model Interpretability and Explainability

XAI Techniques

Global vs Local Interpretability

Global interpretability

- Explains overall model behavior
- Identifies important features

Example:

Understanding fraud risk drivers

Local interpretability

- Explains individual predictions
- Provide case-specific reasoning

Example:

Explaining why a specific transaction was flagged

4.2. Model Interpretability and Explainability

XAI Techniques

Global vs Local Interpretability

Method	Local	Global
SHAP	✓	✓
LIME	✓	X
PDP	X	✓
ALE	X	✓

4.3. Model Interpretability and Explainability

SHAP

SHAP – The current Industry Standard

- Based on cooperative game theory.
 - Quantifies feature contribution to predictions.
- Provides both **global** and **local** explanations.
- Consistent and theoretically grounded.
- Widely adopted across industry and academia.

The advantages:

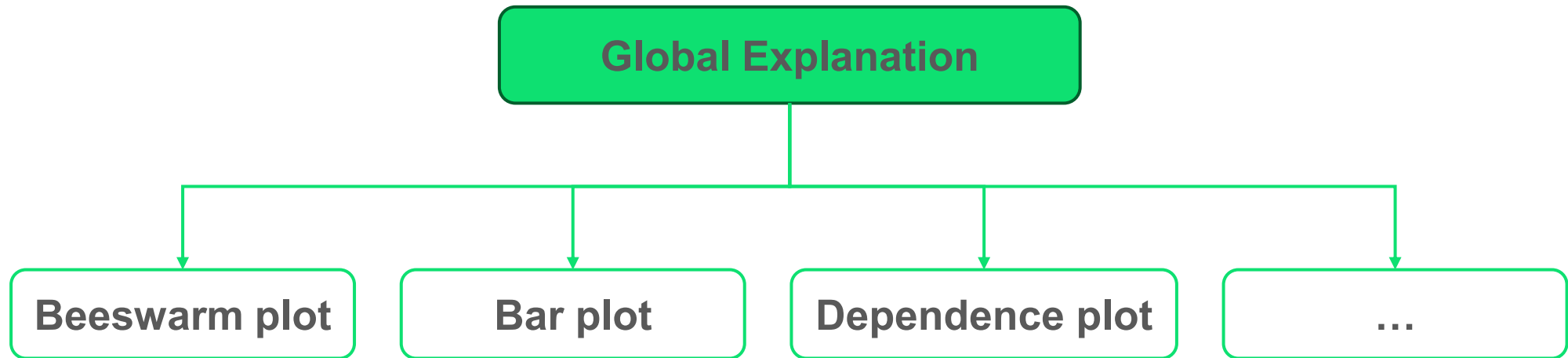
- **Model-agnostic**
- **Handles complex interactions**
- **Works with most traditional ML models**



<https://shap.readthedocs.io/en/latest/>

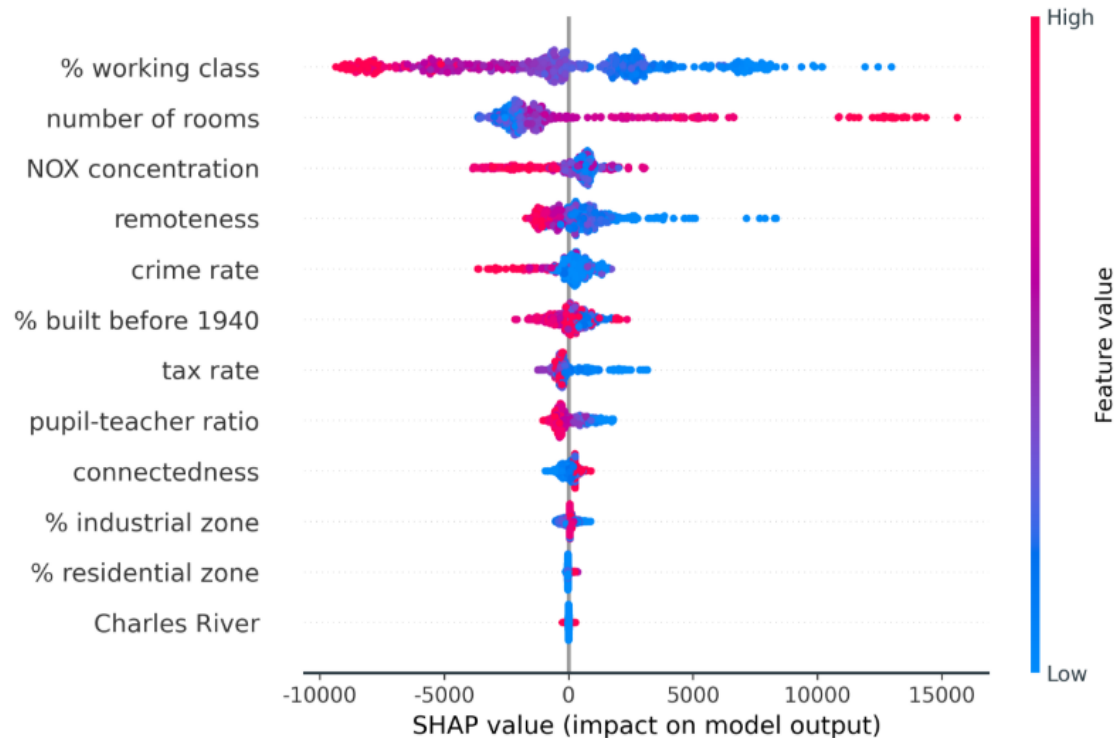
4.3. Data Quality challenges

SHAP – Global Explanation



4.3.1. Model Interpretability and Explainability

SHAP – Beeswarm plot

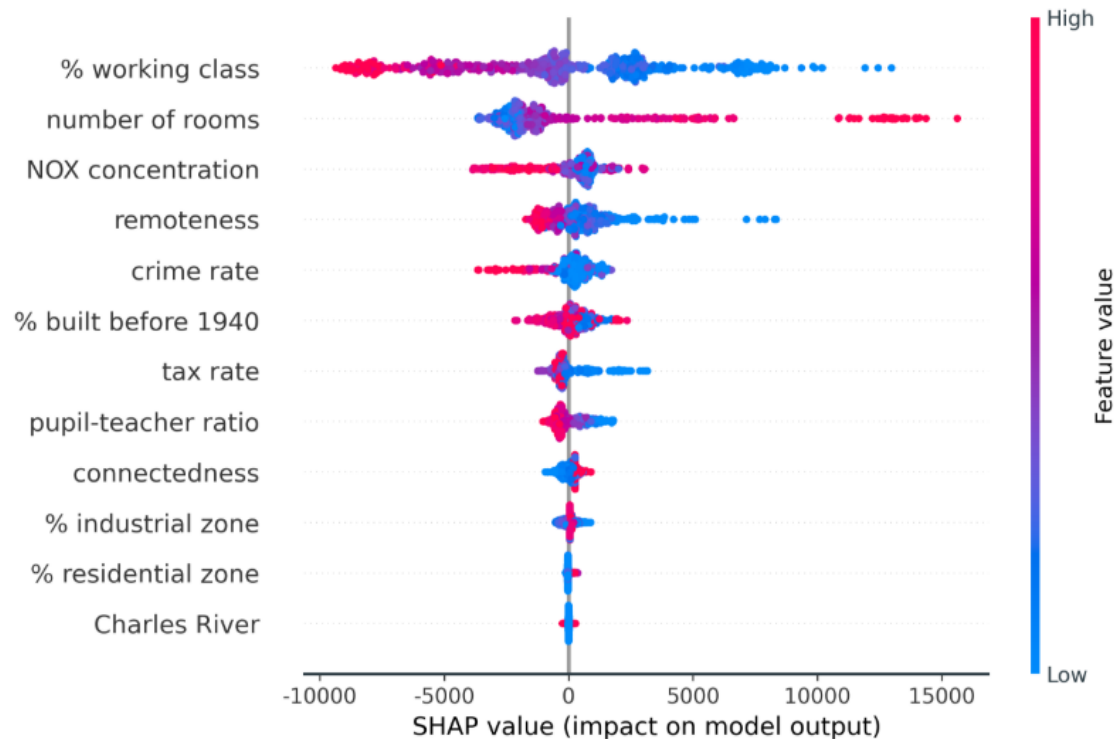


A SHAP **beeswarm plot** is a compact way to visualize the distribution of SHAP values for each feature, showing both the magnitude and direction of impact on model predictions.

- Features are ranked by overall importance (top = most influential)
- Each point is one observation
- X-axis (SHAP value) shows how much a feature pushes the prediction up (right) or down (left).
- Color indicates the feature value (red = high, blue = low).

4.3.1. Model Interpretability and Explainability

SHAP – Beeswarm plot



Main insights:

Working class is the most influential feature

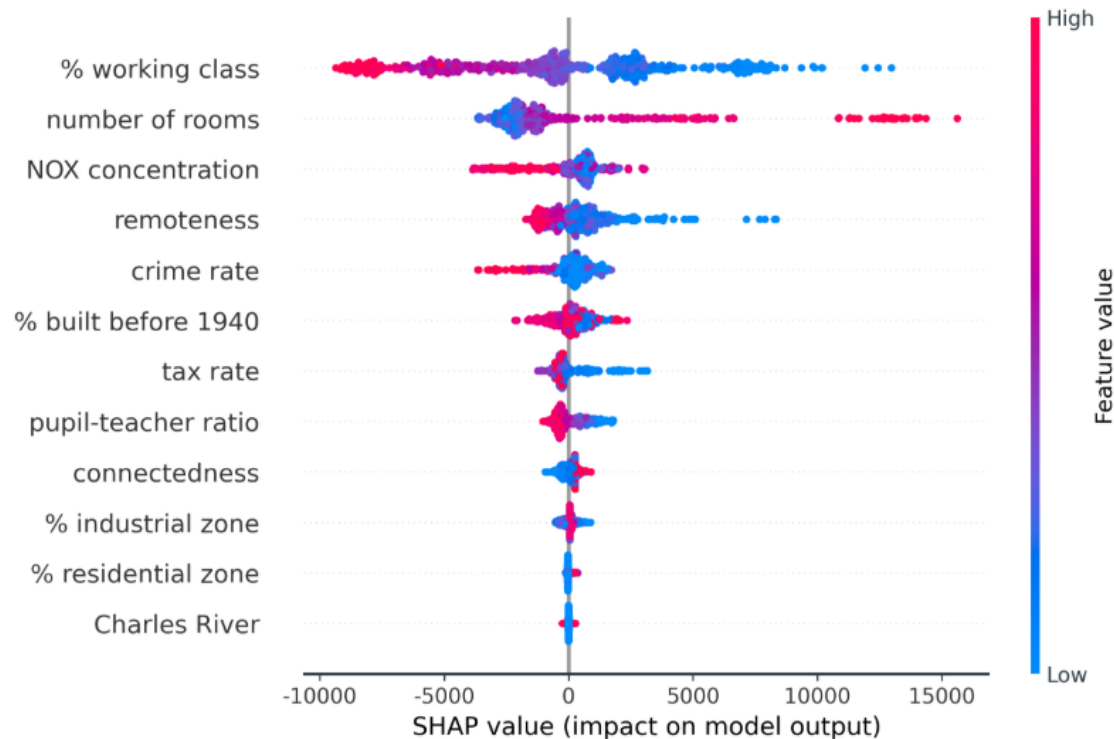
- It has the widest SHAP spread, indicating the largest impact on predictions.
 - **High values** (% working class, red points) are mostly associated with **negative SHAP values**.
 - **Low values** (blue points) tend to increase the prediction.



Interpretation: neighborhoods with a larger working-class population tend to reduce the predicted outcome, the house price (Dataset used: Boston Housing Dataset).

4.3.1. Model Interpretability and Explainability

SHAP – Beeswarm plot



Main insights:

Number of rooms strongly increases predictions

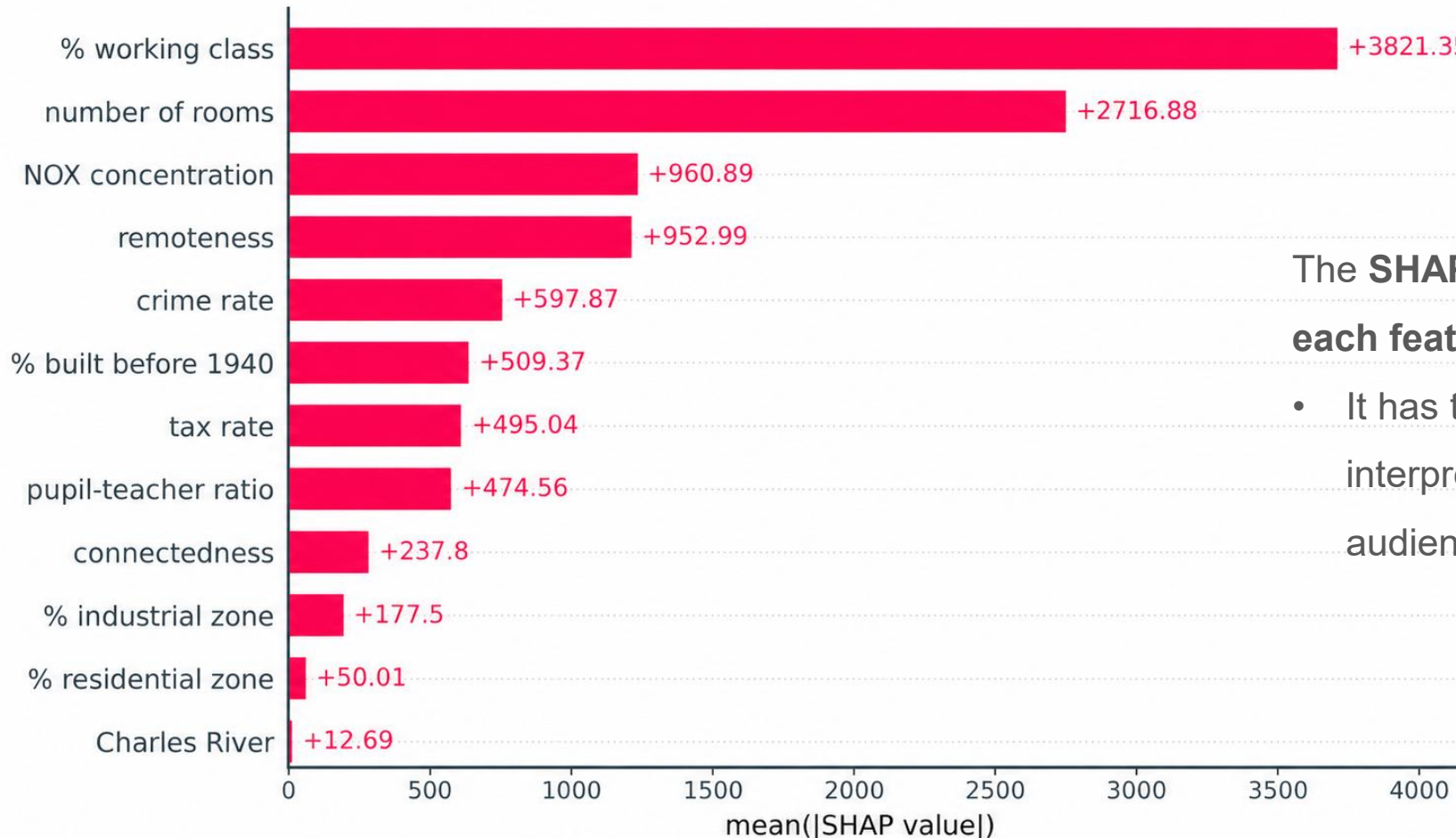
- High values (red) are concentrated on the positive side.
- Low values (blue) are mostly negative.
- More rooms consistently push predictions upward.

Connectedness, industrial zone %, residential zone %, and Charles River proximity contribute little

- Their SHAP values are tightly centered around zero.
- The model relies on them much less than the top-ranked features.

4.3.2. Model Interpretability and Explainability

SHAP – Bar Plot

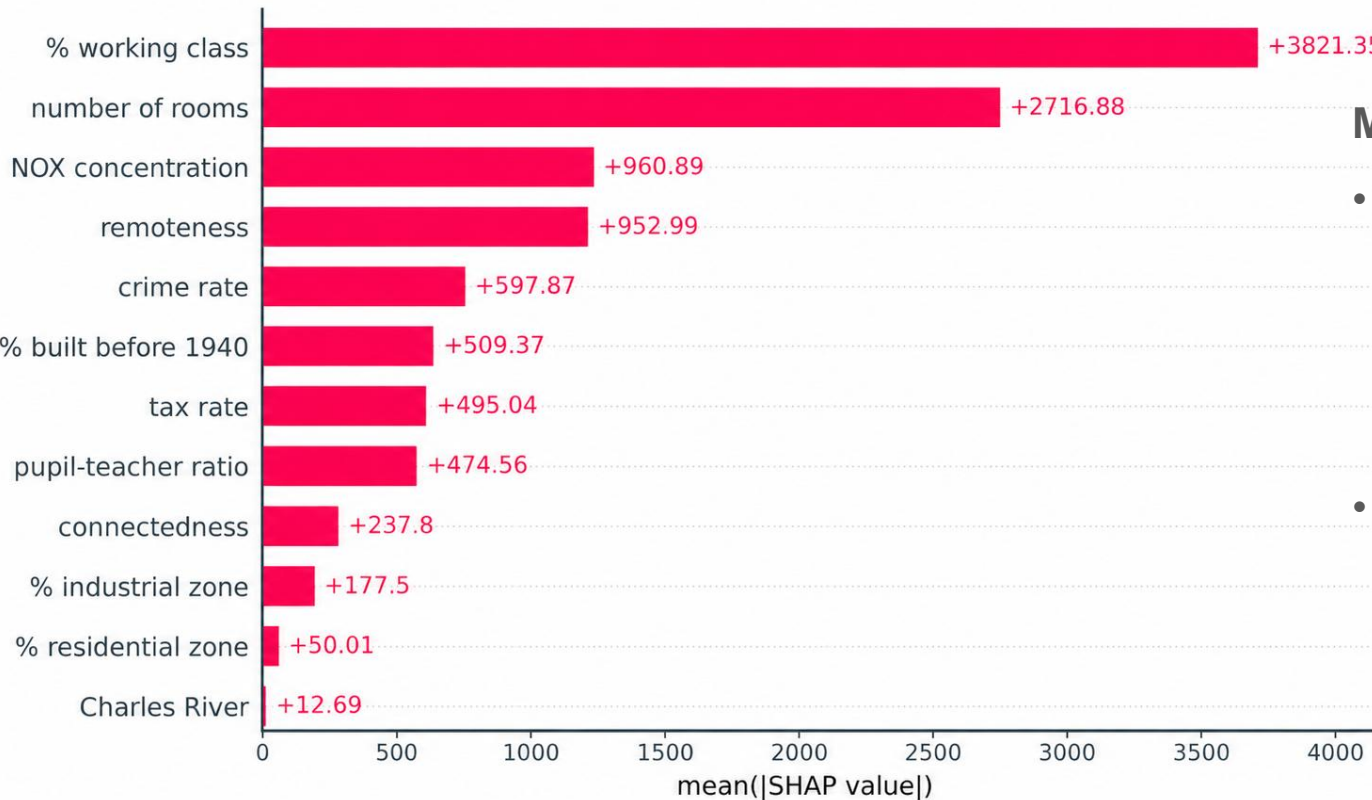


The **SHAP** bar plot shows how important **each feature is overall** (magnitude only).

- It has the advantage of being of very easy interpretation even for non – technical audiences compared to beeswarm.

4.3.2. Model Interpretability and Explainability

SHAP – Bar Plot

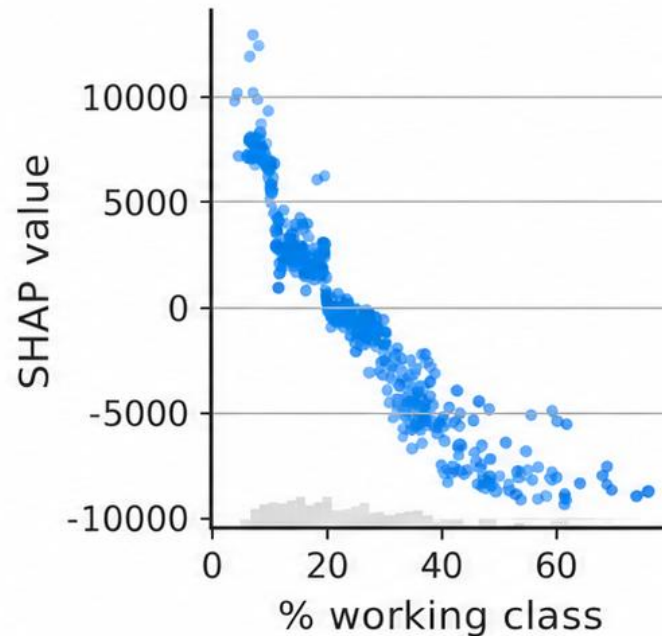


Main insights:

- The **% working class** is the most influential feature, but it does **not** tell us whether higher values increase or decrease house prices; the beeswarm plot provides that information.
- A mean(|SHAP|) of **3821.35** means that changing the "% working class" feature contributes, on average, about **\$3,821** to the prediction relative to the baseline prediction (Be aware that the sign **does not mean the feature increases the prediction.**)

4.3.3. Model Interpretability and Explainability

SHAP – Dependence Plots



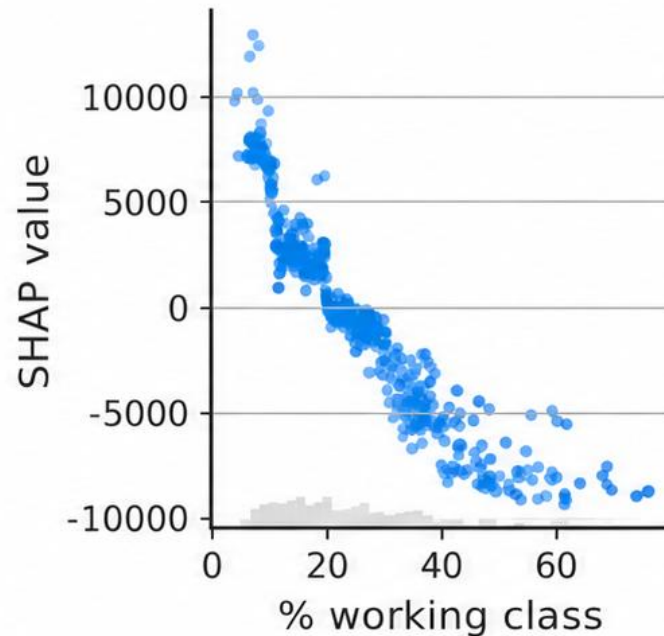
The **SHAP** dependence plot shows how the value of a feature affects the model's prediction across all observations.

It helps to see:

- Whether a feature increases or decreases predictions.
- Potential feature interactions with other variables.

4.3.3. Model Interpretability and Explainability

SHAP – Dependence Plots



Main insights:

Strong negative relationship

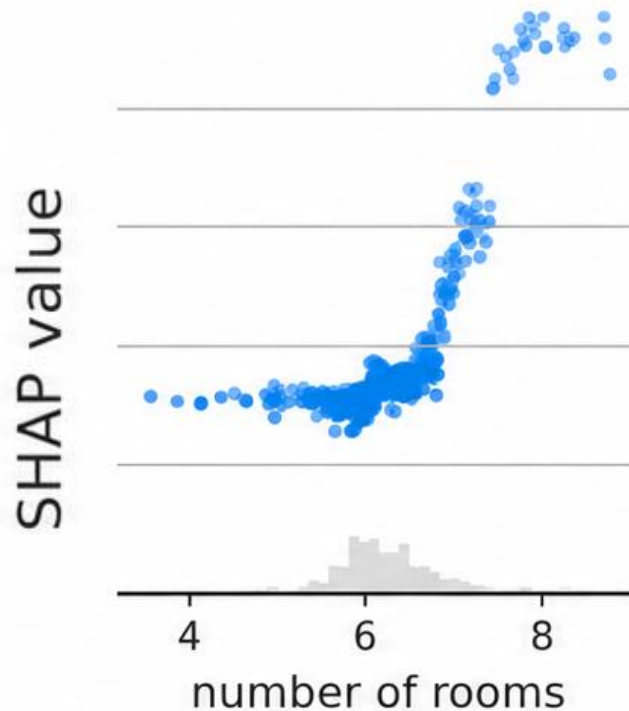
- As **% working class increases**, the SHAP value decreases.
- This means higher values of % working class push the model prediction downward, while lower values push it upward

The feature is highly influential

- SHAP values range roughly from +12,000 to -10,000, indicating this feature has a large impact on the prediction.
- Large absolute SHAP values suggest the model relies heavily on this variable.

4.3.3. Model Interpretability and Explainability

SHAP – Dependence Plots



Main insights:

Strong positive relationship

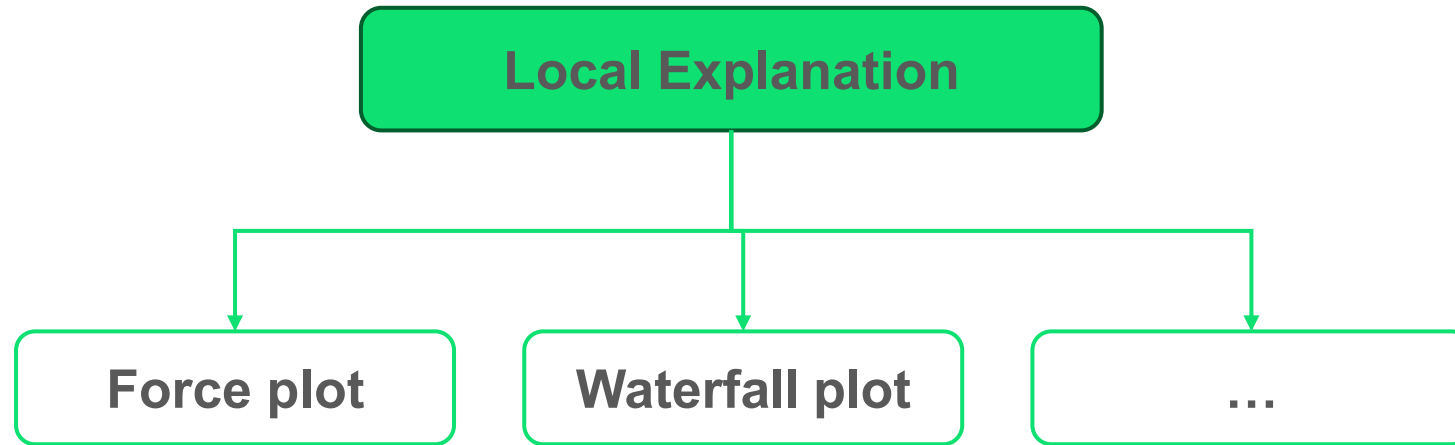
- As the **number of rooms** increases, the SHAP value increases.
- More rooms generally push the model prediction higher.

Highly non-linear effect

- The relationship is not linear.
 - Between about 4 and 6 rooms, the effect is relatively small
 - Around 6.5–7 rooms, the SHAP value rises sharply.

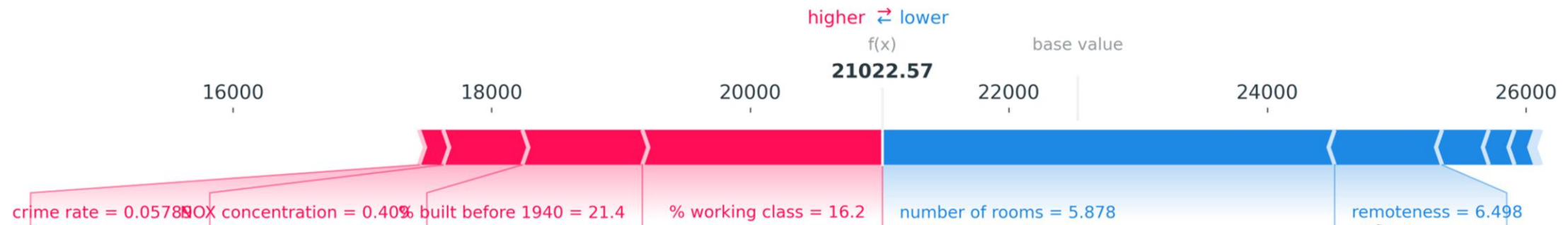
4.3. Data Quality challenges

SHAP – Local Explanation



4.3.4. Model Interpretability and Explainability

SHAP – Force Plots

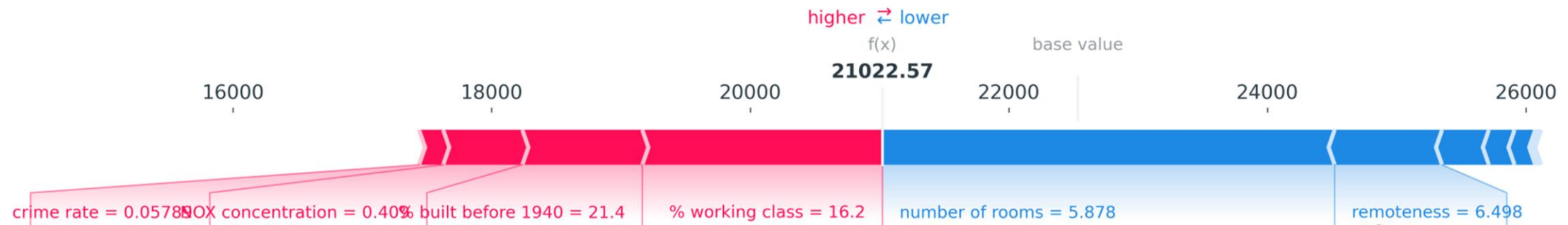


A **SHAP force plot** explains **how individual features contribute to a single prediction** by showing how they push the prediction away from the model's baseline (average) value.

- Red features push the prediction higher.
- Blue features push the prediction lower.
- The size of each segment indicates the strength of the contribution.
- The final result is the model's prediction after combining all feature effects.

4.3.4. Model Interpretability and Explainability

SHAP – Force Plots



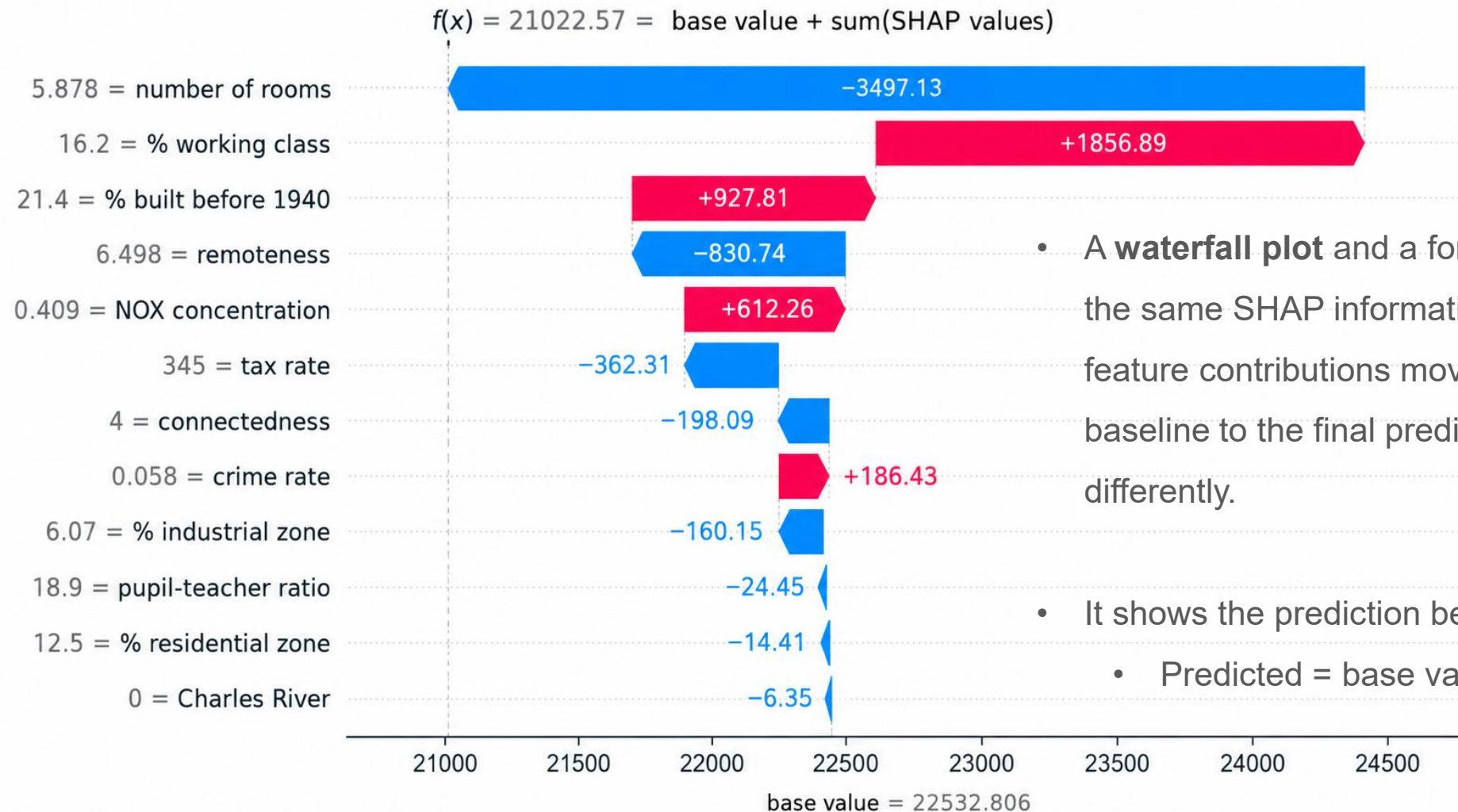
It explains why the model predicted a house price of \$21,022.57 instead of the baseline value of \$22,533.

Main insights:

- The **% working class**, the **% built before 1940**, the **nox concentration** and the **crime rate** associated to this specific house is pushing the prediction up.
- The **number of rooms** and the **remoteness** values associated to this observation are pushing the prediction down.
- For this house, the **number of rooms** had a stronger influence on the prediction than **remoteness**, contributing more to lowering the predicted value.

4.3.5. Model Interpretability and Explainability

SHAP – Waterfall plot



- A **waterfall plot** and a force plot contain essentially the same SHAP information (they both explain how feature contributions move a prediction from the baseline to the final prediction) but they present it differently.

- It shows the prediction being built step by step
 - Predicted = base value + sum (SHAP values)

5.2 Model Interpretability and Explainability

SHAP – RECAP

Explains how features influence model predictions.

Provides both **global** and **local** explanations.

Key plots:

- **Bar Plot** → Feature importance
- **Beeswarm Plot** → Importance + impact direction
- **Dependence Plot** → Feature effect on predictions
- **Force/Waterfall Plot** → Single prediction explanation

5.2 Model Interpretability and Explainability

RECAP

Importance of XAI (Explainable AI)

- **Builds Trust** – Helps users understand and trust model predictions.
- **Improves Transparency** – Reveals how and why decisions are made.
- **Supports Debugging** – Identifies biases, errors, and unexpected model behavior.
- **Ensures Compliance** – Helps meet regulatory and ethical requirements.
- **Enhances Decision-Making** – Provides insights that support human judgment.

XAI makes AI systems more transparent, trustworthy, and accountable.

Obrigada!

Thank you!

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